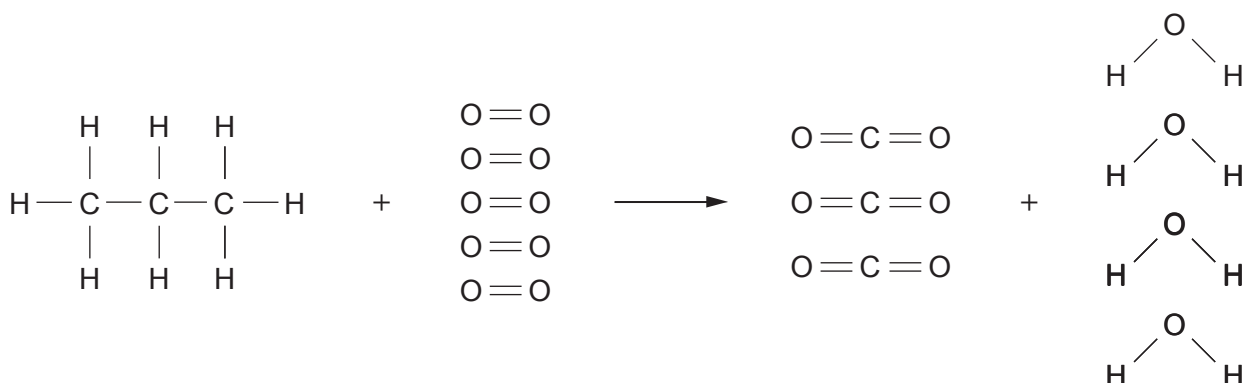


3. (a) Propane gas is commonly used as a fuel. The burning of propane gas is represented by the following equation.



The relative amounts of energy needed to break some of the bonds are given in the table.

| Bond | Bond energy (kJ) |
|------|------------------|
| C—H  | 413              |
| C—C  | 348              |

- (i) The total energy needed to break all of the bonds in the reactants is 6490 kJ.

Calculate the energy needed to break **one** O = O bond.

[3]

Energy needed = ..... kJ



(ii) The energy released when all of the bonds in the products are formed is 8 542 kJ.

Tick (✓) the correct overall energy change for this reaction.

[1]

2 052 kJ taken in

☐

15 032 kJ taken in

☐

15 032 kJ given out

☐

2 052 kJ given out

☐

- (b) Propane gas is commonly used in camping stoves.



The photograph shows a camping stove being used to heat 750 g of water.

When heated for 10 minutes, 220 kJ of energy was transferred to the water.

Use the following equation to calculate the temperature rise of the water.

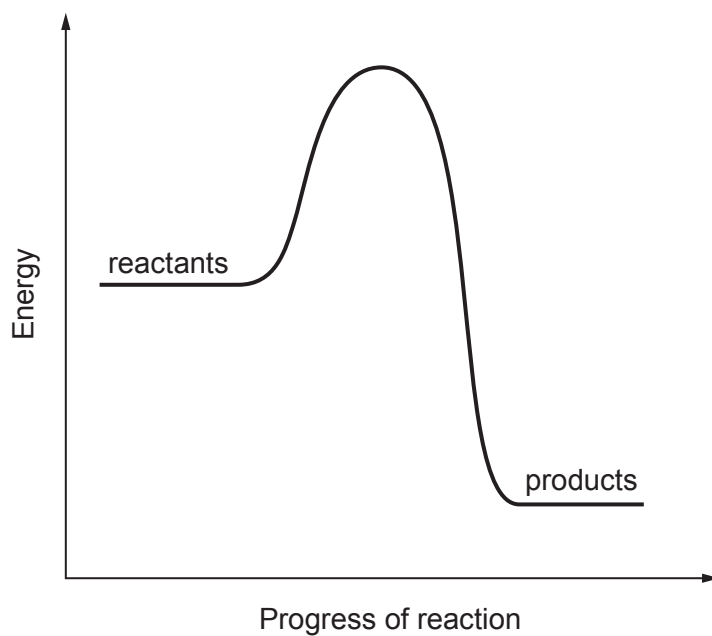
[1]

$$\text{temperature rise (}^{\circ}\text{C)} = \frac{\text{energy transferred (J)}}{4.2 \times \text{mass of water (g)}}$$

Temperature rise = .....  $^{\circ}\text{C}$



(c) The energy profile diagram for the combustion of propane is given below.



- (i) Draw an **arrow** ( $\uparrow$ ) **on the diagram** to show the activation energy.

Label this **A**.

[1]

- (ii) Draw an **arrow** ( $\uparrow$ ) **on the diagram** to show the overall energy change.

Label this **B**.

[1]

